

ROSEBURG, OR, USA

WMO: 726904

Lat: 43.239N

Lon: 123.355W

Elev: 160

StdP: 99.42

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 24231

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.3	-0.9	-5.9	2.3	1.6	-3.0	3.0	4.3	7.0	10.2	5.7	10.2	0.5	0	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	34.7	19.5	32.7	19.0	30.9	18.4	20.7	32.1	19.9	30.8	19.0	29.2	4.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	11.9	24.3	15.7	11.4	23.6	14.8	10.7	22.4	60.0	32.1	56.9	30.9	54.1	29.2	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.5	4.8	DB	-5.1	39.1	3.1	2.1	-7.4	40.6	-9.3	41.8	-11.0	43.0	-13.3	44.5	(4)
(5)			WB	-5.7	22.5	3.2	1.1	-8.0	23.3	-9.9	23.9	-11.6	24.5	-14.0	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	6.4	7.4	9.4	11.5	15.0	18.2	22.0	22.0	18.9	13.3	8.7	5.7	(6)	
	DBStd	6.54	3.17	2.82	3.06	3.37	3.55	3.46	2.96	2.67	3.11	3.09	3.36	3.51	(7)	
	HDD10.0	478	118	79	49	21	2	0	0	0	0	7	62	138	(8)	
	HDD18.3	2191	371	306	278	207	114	42	4	3	30	157	289	391	(9)	
	CDD10.0	1666	6	8	30	66	158	247	372	372	267	111	24	6	(10)	
	CDD18.3	337	0	0	0	2	12	40	118	117	46	3	0	0	(11)	
	CDH23.3	4095	0	0	2	54	214	519	1355	1291	606	54	0	0	(12)	
	CDH26.7	1752	0	0	0	14	71	199	628	588	241	11	0	0	(13)	
(14) Wind	WSAvg	1.5	0.9	1.1	1.3	1.6	1.8	2.2	2.3	2.0	1.5	1.1	1.0	1.0	(14)	
(15) Precipitation	PrecAvg	840	132	93	91	64	47	21	9	14	25	57	139	149	(15)	
	PrecMax	1529	288	248	178	179	161	68	76	84	94	135	404	401	(16)	
	PrecMin	551	15	25	11	15	7	0	0	0	0	2	28	21	(17)	
	PrecStd	187	72	56	40	36	35	17	14	19	22	34	77	86	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	17.9	22.8	28.8	32.2	35.2	37.9	37.3	34.9	27.9	21.0	16.7	(19)	
		MCWB	11.8	11.8	13.7	16.4	18.4	20.1	20.7	20.1	18.4	16.0	15.6	12.4	(20)	
	2%	DB	14.0	15.5	20.0	24.1	28.5	31.5	34.7	34.1	32.2	24.4	17.4	14.1	(21)	
		MCWB	10.8	10.8	12.3	14.5	16.9	18.8	19.7	19.8	18.1	14.8	12.8	11.3	(22)	
	5%	DB	12.5	13.7	17.5	21.0	25.6	28.7	32.5	32.3	29.5	21.8	15.3	12.4	(23)	
		MCWB	10.0	9.7	11.5	13.1	15.5	17.3	19.1	18.9	17.3	14.3	12.1	10.3	(24)	
	10%	DB	11.1	12.1	15.0	17.9	22.6	26.2	30.2	30.1	27.1	18.8	13.5	10.9	(25)	
		MCWB	9.1	9.1	10.4	11.6	14.4	16.3	18.3	18.3	16.5	13.4	11.0	9.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	13.1	14.6	17.2	19.6	21.1	22.3	21.7	20.4	17.6	15.8	13.4	(27)	
		MCDB	15.2	16.2	20.8	26.9	31.3	32.8	34.1	34.0	31.2	24.9	19.5	15.3	(28)	
	2%	WB	11.6	11.6	13.2	15.1	17.6	19.3	20.7	20.6	19.1	16.2	14.0	11.9	(29)	
		MCDB	13.3	14.2	18.0	22.6	27.0	30.0	32.1	32.0	29.1	22.1	16.3	13.4	(30)	
	5%	WB	10.4	10.4	12.0	13.6	16.1	18.0	19.8	19.7	18.1	15.1	12.5	10.5	(31)	
		MCDB	12.2	12.8	16.2	19.8	24.0	27.3	30.8	30.7	27.3	19.8	14.4	12.2	(32)	
	10%	WB	9.2	9.4	10.9	12.2	14.8	16.8	18.9	18.8	17.0	14.1	11.4	9.0	(33)	
		MCDB	10.8	11.7	14.2	17.2	21.6	24.9	29.1	28.8	25.3	18.0	13.2	10.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	8.1	9.8	11.0	12.0	12.8	15.1	15.4	14.7	11.0	7.0	5.8	(35)	
		MCDBR	8.7	10.9	14.6	17.4	18.2	17.8	18.6	19.0	19.7	16.3	9.7	7.7	(36)	
	5% WB	MCWBR	5.5	6.5	7.6	8.4	8.0	7.1	6.8	7.0	7.9	7.8	5.7	5.0	(37)	
		MCWBR	7.7	9.1	12.8	15.7	16.6	16.3	17.6	17.4	17.2	13.1	8.1	7.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.310	0.316	0.316	0.340	0.345	0.348	0.344	0.345	0.330	0.323	0.309	0.317	(40)	
		taud	2.564	2.553	2.572	2.474	2.486	2.488	2.514	2.525	2.550	2.574	2.585	2.531	(41)	
	Edn at Noon	Edn at Noon	834	890	933	927	927	921	921	911	904	868	829	786	(42)	
		Edh at Noon	67	79	88	104	106	106	103	98	89	77	65	64	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.52	2.33	3.38	4.73	6.03	6.79	7.50	6.52	4.96	3.05	1.68	1.22	(44)	
	RadStd	RadStd	0.21	0.31	0.39	0.41	0.52	0.51	0.38	0.34	0.37	0.30	0.20	0.20	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+132	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+10

Nomenclature: See separate page